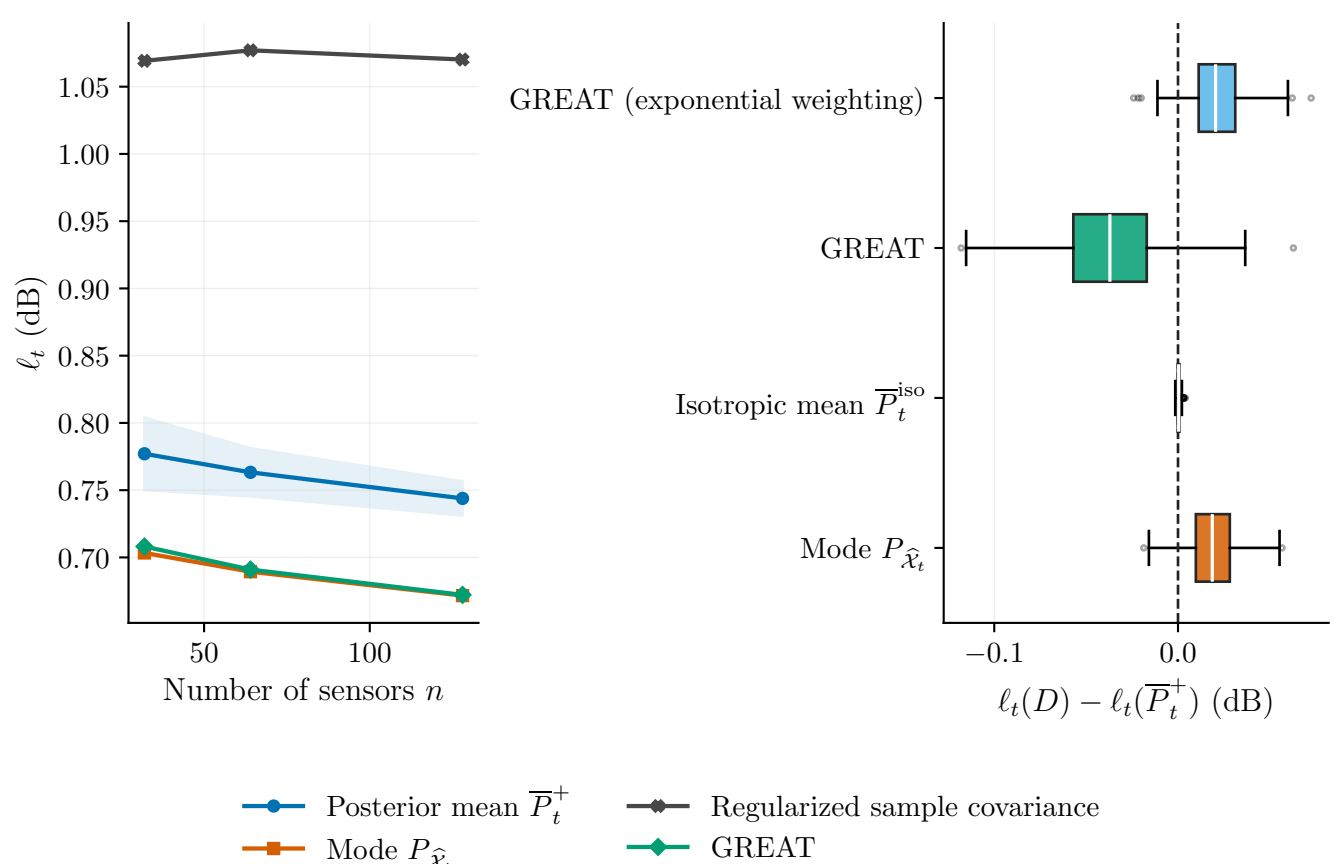
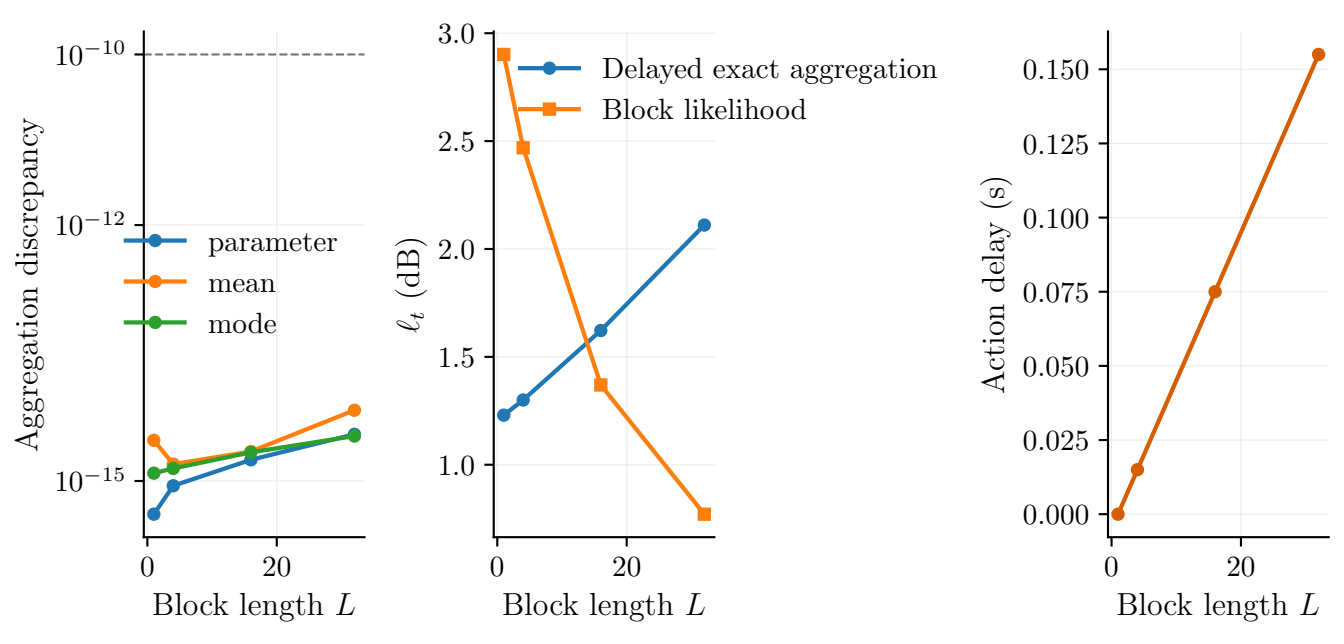
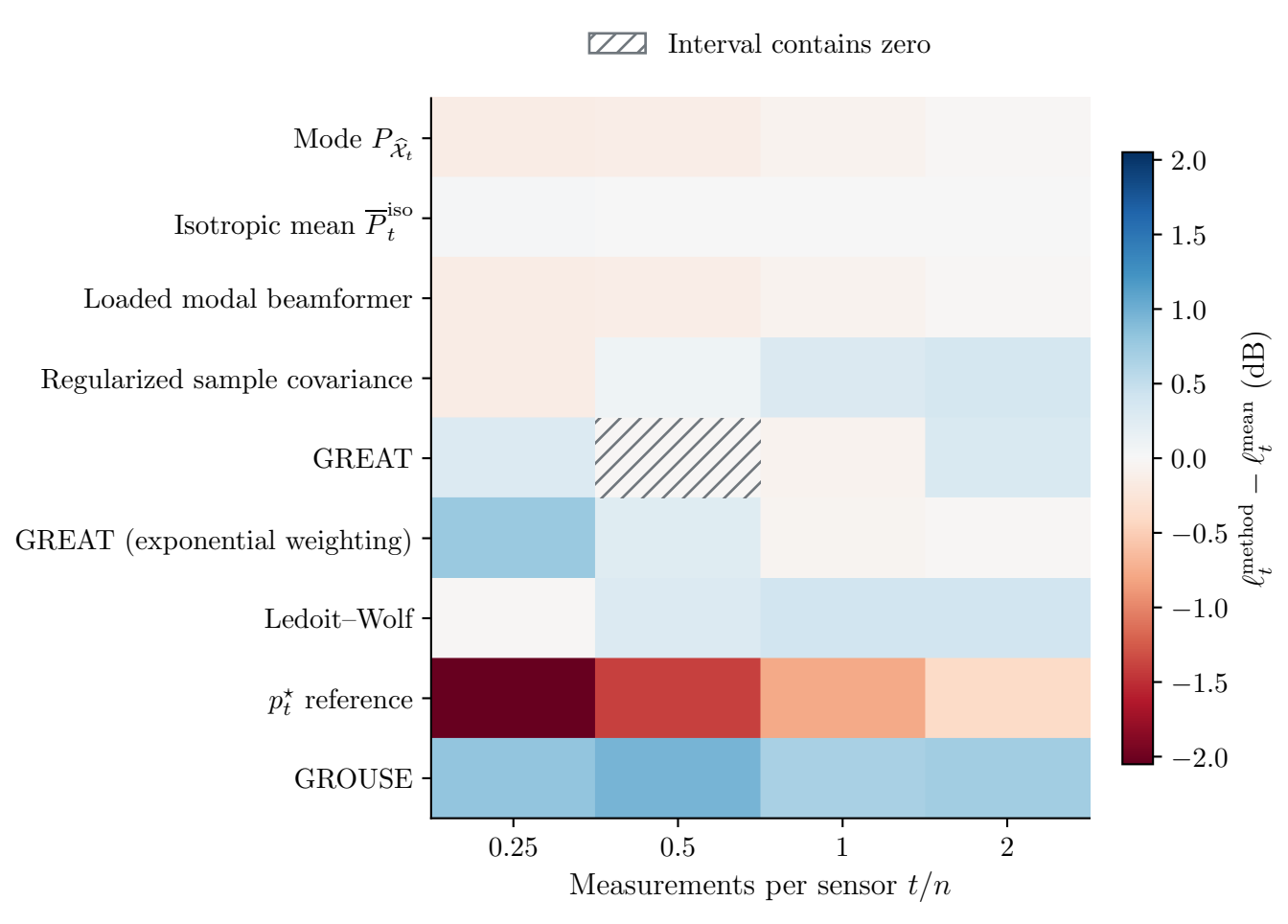
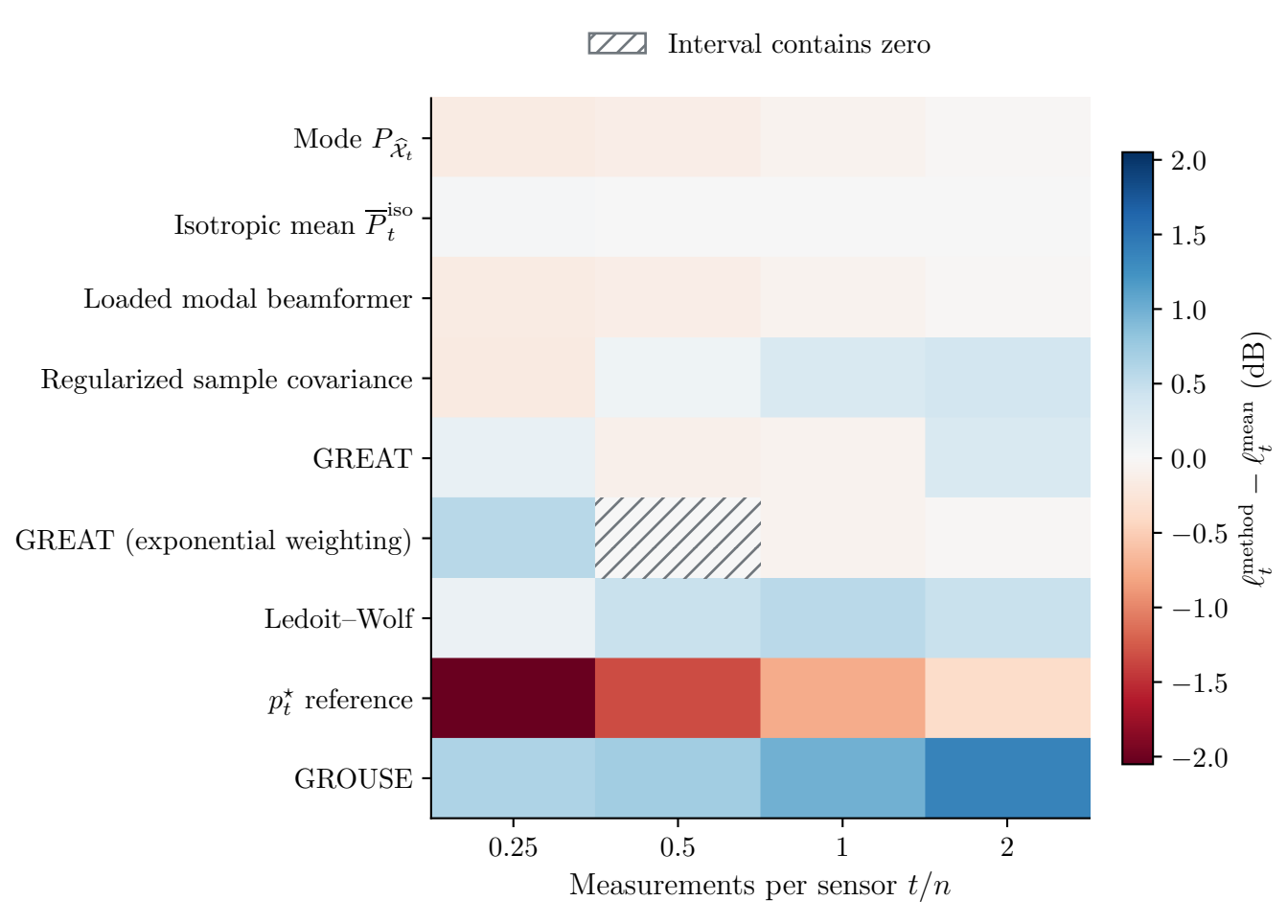
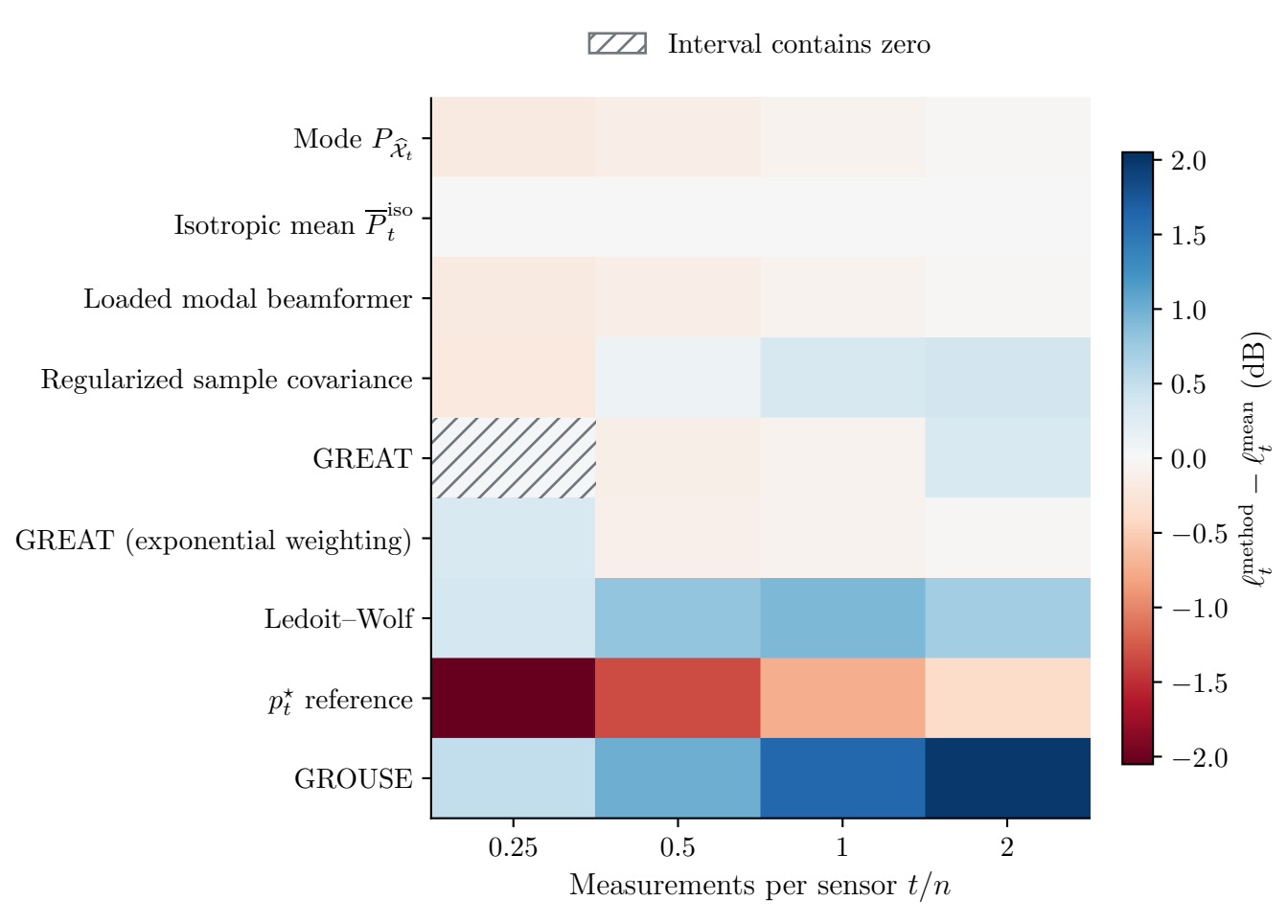
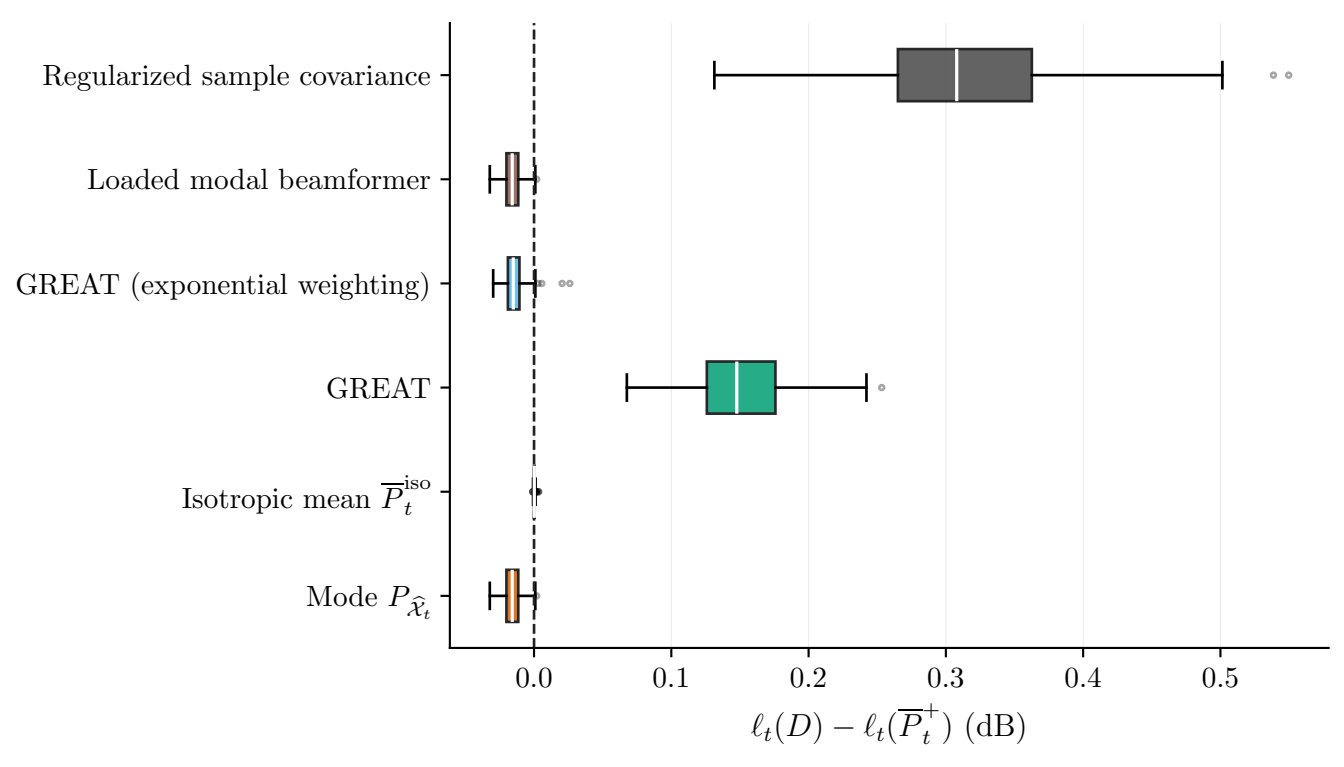

1.05
1.00
0.95
0.90
0.85
0.80
0.75
0.70
ℓ_t (dB)
50
100
Number of sensors n
GREAT (exponential weighting)
GREAT
Isotropic mean $\overline{P}_t^{\text{iso}}$
Mode $P_{\widehat{\mathcal{X}}_t}$
−0.1
0.0
$\ell_t(D) - \ell_t(\overline{P}_t^+)$ (dB)
Posterior mean $\overline{P}_t^+$
Mode $P_{\widehat{\mathcal{X}}_t}$
Regularized sample covariance
GREAT

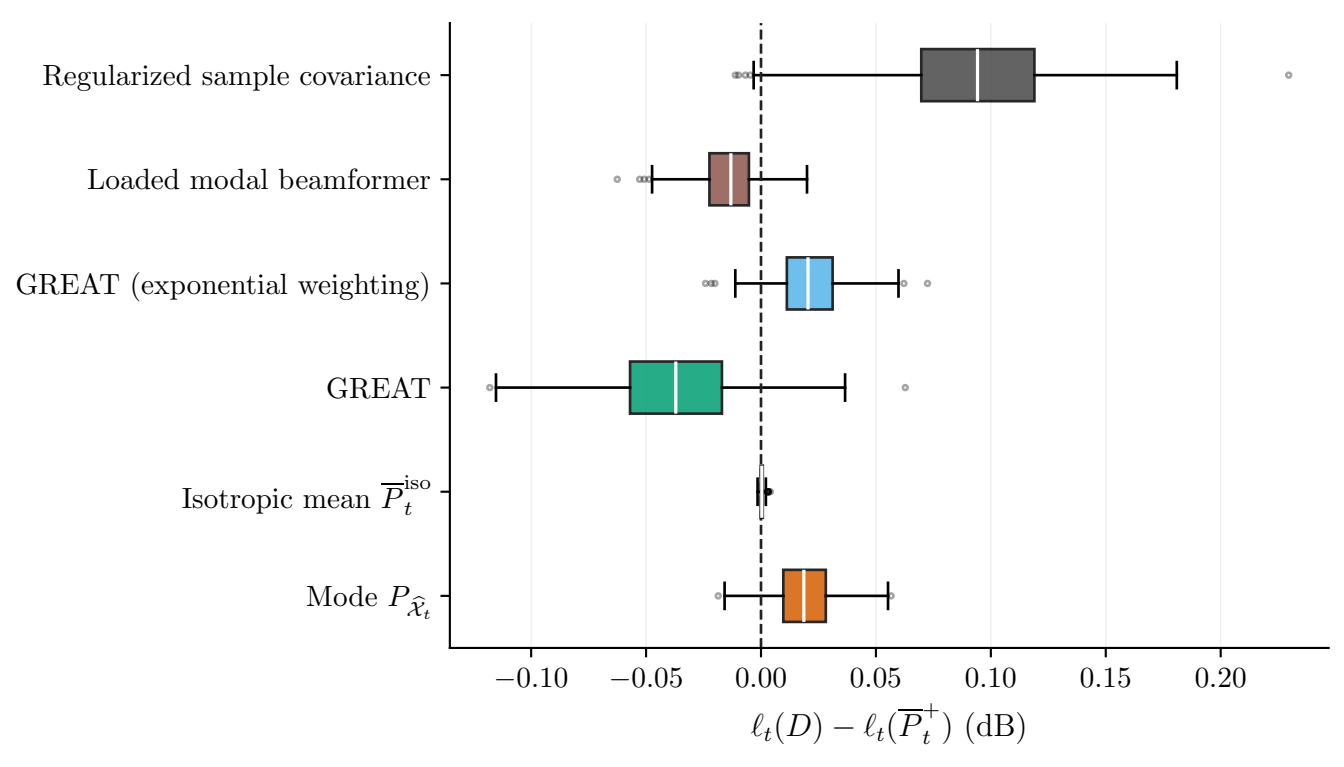

Aggregation discrepancy
10^{-10}
10^{-12}
10^{-15}
parameter
mean
mode
0
20
Block length L
ℓ_t (dB)
3.0
2.5
2.0
1.5
1.0
Delayed exact aggregation
Block likelihood
0
20
Block length L
Action delay (s)
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0
20
Block length L

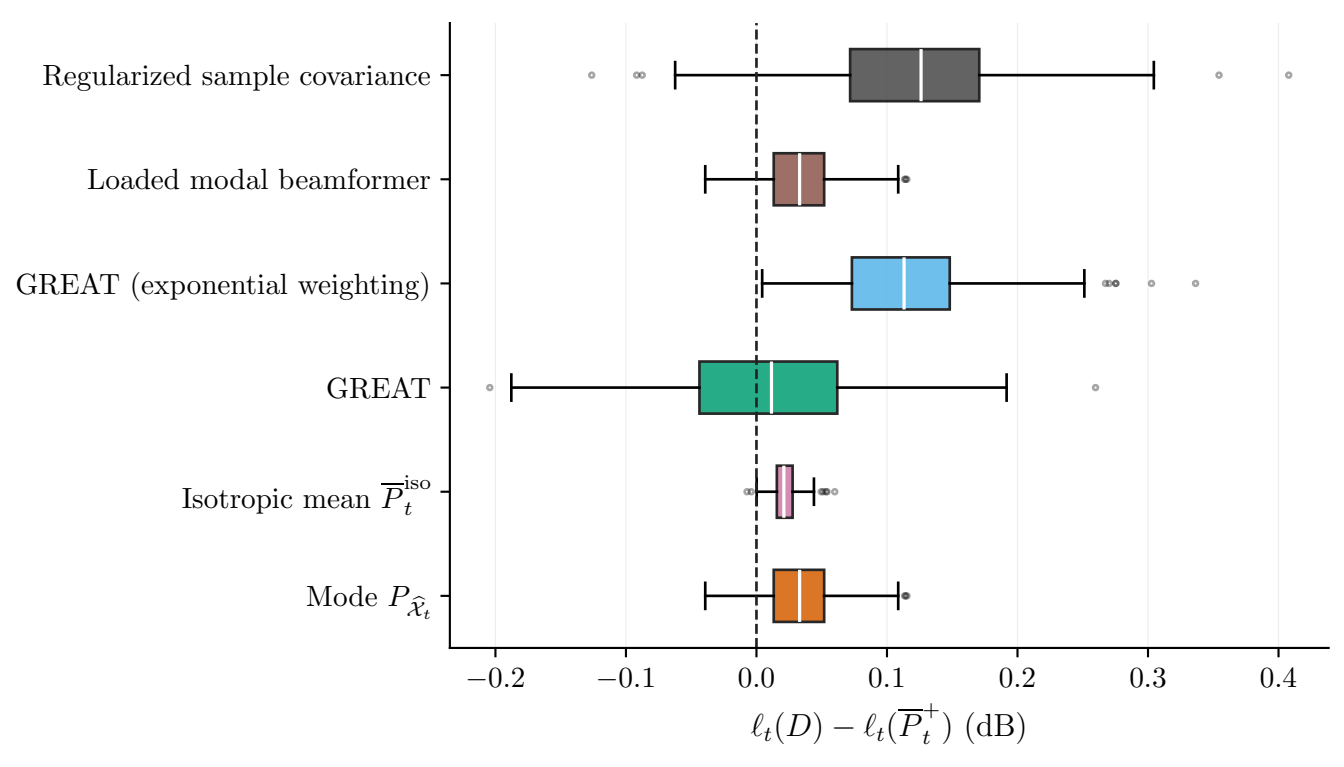

Interval contains zero
Mode $P_{\widehat{\mathcal{X}}_t}$
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Loaded modal beamformer
Regularized sample covariance
GREAT
GREAT (exponential weighting)
Ledoit–Wolf
$p_t^\star$ reference
GROUSE
0.25
0.5
1
2
Measurements per sensor t/n
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
$\ell_t^{\mathrm{method}} - \ell_t^{\mathrm{mean}}$ (dB)

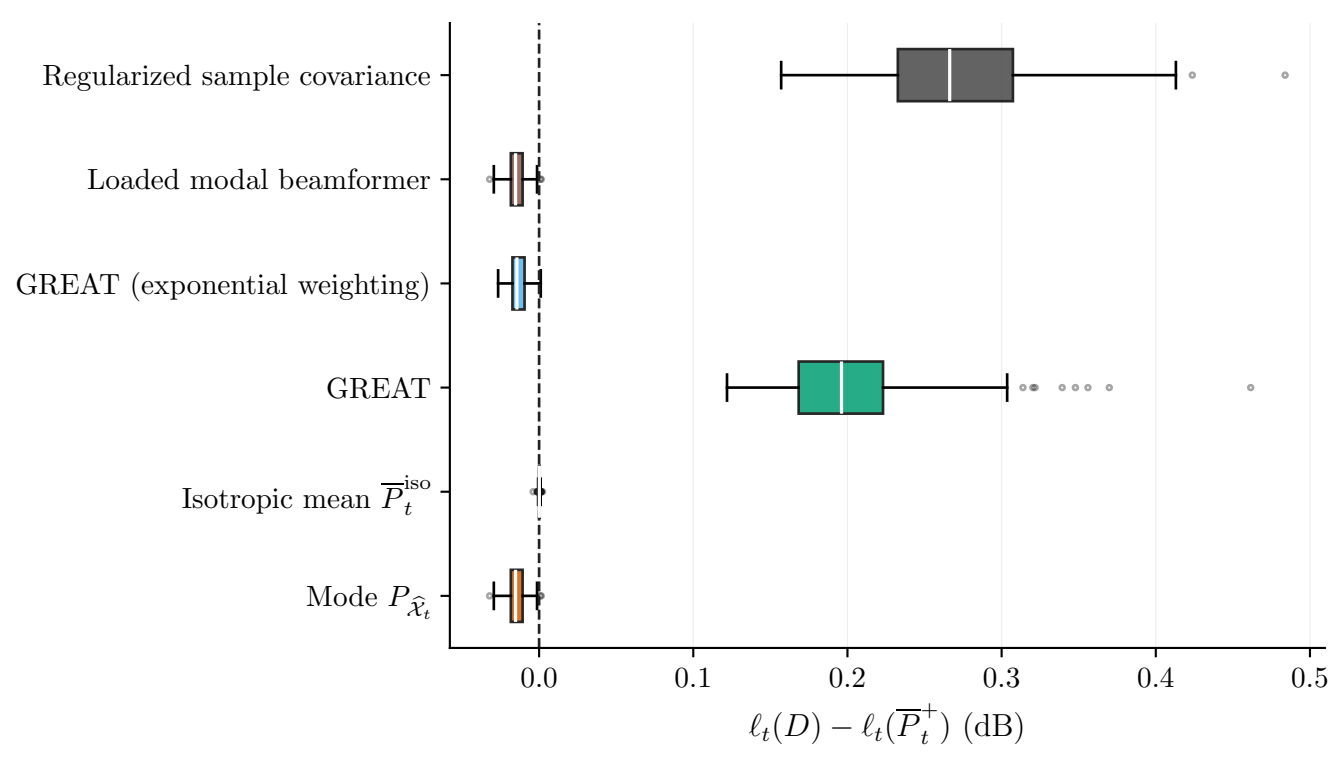

Interval contains zero
Mode $P_{\widehat{\mathcal{X}}_t}$
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Loaded modal beamformer
Regularized sample covariance
GREAT
GREAT (exponential weighting)
Ledoit–Wolf
$p_t^\star$ reference
GROUSE
0.25
0.5
1
2
Measurements per sensor t/n
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
$\ell_t^{\mathrm{method}} - \ell_t^{\mathrm{mean}}$ (dB)

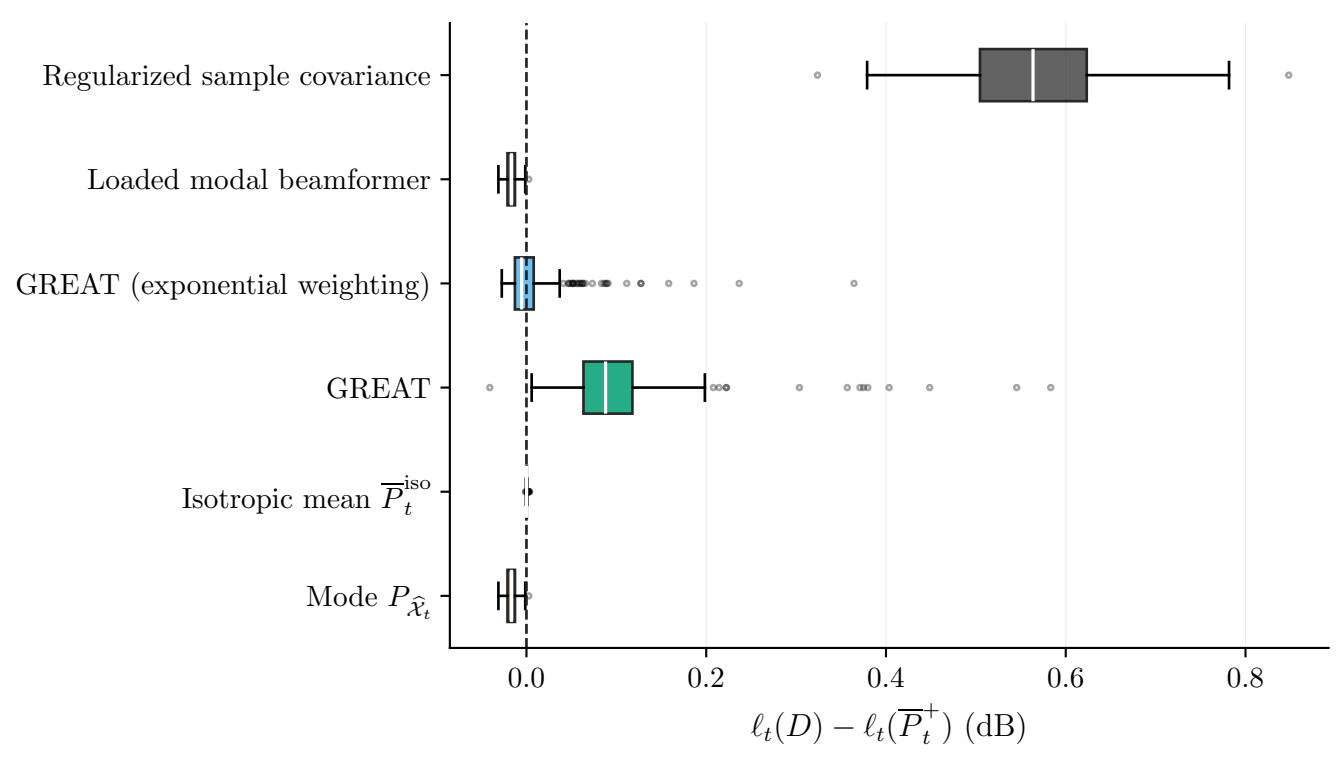

Interval contains zero
Mode $P_{\widehat{\mathcal{X}}_t}$
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Loaded modal beamformer
Regularized sample covariance
GREAT
GREAT (exponential weighting)
Ledoit–Wolf
$p_t^\star$ reference
GROUSE
0.25
0.5
1
2
Measurements per sensor t/n
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
$\ell_t^{\mathrm{method}} - \ell_t^{\mathrm{mean}}$ (dB)

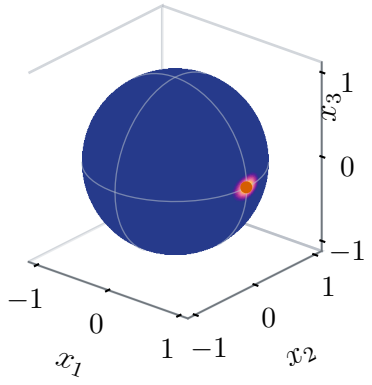

Regularized sample covariance
Loaded modal beamformer
GREAT (exponential weighting)
GREAT
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Mode $P_{\widehat{\mathcal{X}}_t}$
0.0
0.1
0.2
0.3
0.4
0.5
$\ell_t(D) - \ell_t(\overline{P}_t^{+})$ (dB)

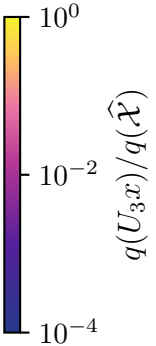

Regularized sample covariance
Loaded modal beamformer
GREAT (exponential weighting)
GREAT
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Mode $P_{\widehat{\mathcal{X}}_t}$
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
$\ell_t(D) - \ell_t(\overline{P}_t^{+})$ (dB)

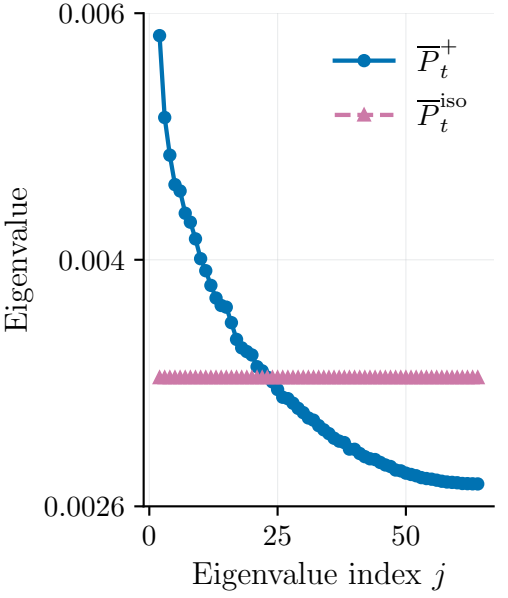

Regularized sample covariance
Loaded modal beamformer
GREAT (exponential weighting)
GREAT
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Mode $P_{\widehat{\mathcal{X}}_t}$
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
$\ell_t(D) - \ell_t(\overline{P}_t^{+})$ (dB)

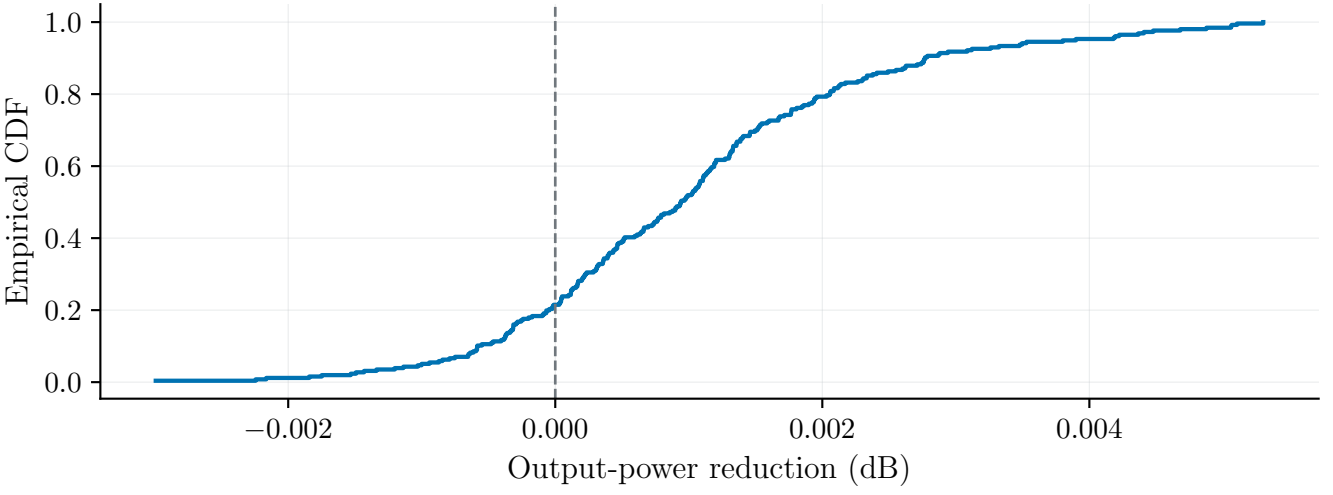

Regularized sample covariance
Loaded modal beamformer
GREAT (exponential weighting)
GREAT
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Mode $P_{\widehat{\mathcal{X}}_t}$
0.0
0.1
0.2
0.3
0.4
0.5
$\ell_t(D) - \ell_t(\overline{P}_t^{+})$ (dB)


Regularized sample covariance
Loaded modal beamformer
GREAT (exponential weighting)
GREAT
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Mode $P_{\widehat{\mathcal{X}}_t}$
0.0
0.2
0.4
0.6
0.8
$\ell_t(D) - \ell_t(\overline{P}_t^{+})$ (dB)


x_3
1
0
−1
−1
0
1
x_1
−1
0
1
x_2


10^0
10^{-2}
10^{-4}
$q(U_3x)/q(\widehat{\mathcal{X}})$


0.006
0.004
0.0026
Eigenvalue
$\overline{P}_t^{+}$
$\overline{P}_t^{\mathrm{iso}}$
0
25
50
Eigenvalue index j


1.0
0.8
0.6
0.4
0.2
0.0
Empirical CDF
−0.002
0.000
0.002
0.004
Output-power reduction (dB)